

Please do the following experiment independently using a free Google Colab environment.

1. SAM's Image Encoder is a ViT Pre-Trained on a SA-1B(Natural Images). When we run the chest X-ray through it, what would you expect the intermediate representations to look like compared to the natural image? Where in the encoder do you expect the medical-domain specific features emerge at all? Be specific about layers and attention.
2. Extract the patch embeddings from the last 4 layers of SAM for both image sets. using these representations, answer these does the encoder separates x-ray from MRIs? show in numbers and visuals on this. Then identify which layer produces the clearest separation ?
3. You ablate (zero-out) the attention head, assume k in a layer L of the encoder, observe the dice score on MRI drops by $x\%$ but x-rays get unaffected. propose 2 mechanistic hypotheses for why this might happen, how would you test each one ?

Use the below for all the above experiments

Model : SAM ViT-H / ViT-B for compute friendliness.

Datasets: 10-30 chest X-rays (from a public dataset e.g. NIH ChestX-ray14)

10-30 abdominal MRI slices (from a public dataset)

All the Images resized to 1024x1024, no prompts needed